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A Gaia DR3 Catalogue of Stars with Very Low Galactic Rest-Frame Speeds

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ABSTRACT

We present a Gaia DR3 catalogue of stars selected by very low Galactic rest-frame speed, $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$. Using Lindegren et al. (2021) parallax zero-point corrections, Bailer-Jones et al. (2021) photogeometric distances, and Monte Carlo propagation of the Gaia astrometric covariance, we identify a headline Tier A+B sample of 334 stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$ and a broader Tier A+B+C sample of 632 stars with $P > 0.50$. Orbit integrations in the static, axisymmetrized Hunter et al. (2024) Galactic potential map the broader sample onto compact, highly radial model orbits, with median eccentricity 0.961, median pericentre 162 pc, and median apocentre 8.08 kpc; 99.2% retain $e > 0.8$ at the conservative 16th percentile of the per-star orbit posterior. A velocity-stratified matched-control comparison shows that the low-$V_{\rm grf}$ catalogue is the compact-pericentre endpoint of a smooth trend across the observed Gaia DR3 6D velocity distribution, rather than a distinct physical boundary at exactly $25\,{\rm km\,s^{-1}}$. Selection-function weights imply a larger effective count but are sensitive to sparse colour–magnitude cells, and external spectroscopy is available for only a small subset. We therefore present the catalogue as a probabilistic kinematic sample with model-dependent orbit diagnostics and subset-qualified chemical context, not as a complete census or population-membership decomposition.

Keywords: Galaxy: halo — Galaxy: kinematics and dynamics — Galaxy: stellar content — catalogs — stars: kinematics and dynamics

1. INTRODUCTION

Gaia DR3 provides full six-dimensional phase-space information for tens of millions of stars through its astrometry and radial velocities (Gaia Collaboration et al. 2023; Katz et al. 2023). This makes it possible to search the observed Milky Way phase-space distribution for rare kinematic states, not only for previously defined stellar populations or merger debris. Here we focus on one such state: stars observed with unusually low instantaneous Galactic rest-frame speed.

The sample is defined by present-day kinematics, not by inferred origin. We construct a probabilistic low-$V_{\rm grf}$ Gaia DR3 catalogue, quantify the stability of the velocity-threshold membership under measurement uncertainties, and report model-dependent orbit diagnostics in an adopted Milky Way potential. The orbit integrations are used to interpret the observed slow state, but they do not define the catalogue.

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This framing is important because low instantaneous speed can arise in more than one physical context. For wide radial orbits it may mark an apocentric phase; for compact inner-Galaxy orbits it may reflect a different part of the orbit distribution. We therefore compare the low-$V_{\rm grf}$ catalogue with matched Gaia DR3 control samples across higher velocity bands and treat external spectroscopic matches as context for the subset with available chemistry, not as a whole-catalogue population assignment.

Section 2 describes the sample definition, uncertainty propagation, controls, and adopted orbit model. Section 3 presents the empirical catalogue properties and chemical cross-matches. Section 4 gives the model-dependent dynamical results. Section 5 discusses the interpretation and its limits, and Section 6 summarises the conclusions.

2. SAMPLE DEFINITION AND METHODS

2.1. *Gaia DR3 Sample*

The parent source is Gaia DR3 (Gaia Collaboration et al. 2023), restricted to stars with full six-dimensional phase-space information: sky position, parallax, proper motion, and radial velocity. Radial velocities in DR3 are available for approximately 33.8 million stars observed by the Radial Velocity Spectrometer (Katz et al. 2023). This 6D subset has a non-trivial selection function in sky position, magnitude, and colour (Castro-Ginard et al. 2023; Cantat-Gaudin et al. 2023); it is not a neutral slice of the Galaxy, and distant 6D samples are commonly giant-heavy, with incompleteness expected for cool faint dwarfs.

The sample was constructed from a local mirror of the Gaia DR3 source table using the columns needed for a six-dimensional solution: `source_id`, `ra`, `dec`, `parallax`, `pmra`, `pmdec`, `radial_velocity`, the corresponding uncertainties and astrometric correlations, broad-band photometry, `ruwe`, and `duplicated_source`. The equivalent Gaia Archive selection is `radial_velocity IS NOT NULL`, `parallax > 0`, `pmra IS NOT NULL`, `pmdec IS NOT NULL`, and `parallax_over_error > 5`. No RUWE or duplicated-source cut is applied to define the point-estimate catalogue; those flags define the Gold subset below. Heliocentric observables are transformed into Galactocentric coordinates and velocities, and Galactic rest-frame speeds are computed.

The propagated candidate pool contains 2,859 Gaia DR3 6D sources. After the final parallax zero-point, photogeometric-distance, and Solar-frame conventions are applied, 709 sources have corrected point-estimate $V_{\mathrm{grf}} < 25\,\mathrm{km\,s^{-1}}$. Monte Carlo propagation of the observables then defines the catalogue tiers used throughout the paper: Tier A ($P > 0.95$, 214 stars), Tier A+B ($P > 0.84$, 334 stars), and Tier A+B+C ($P > 0.50$, 632 stars). The remaining 77 corrected point-estimate sources below $25\,\mathrm{km\,s^{-1}}$ have $P \leq 0.50$ and are retained as Tier D rather than as part of the headline catalogue.

This is not a volume-complete sample and it is not intended as a complete inventory of all such stars in the Milky Way. It is a Gaia DR3 6D catalogue defined by a simple instantaneous kinematic state and reported with explicit threshold-membership probabilities.

To keep the role of each count unambiguous, Table 2 maps each tier and derived sample to the results it anchors. The **headline catalogue** for the paper is Tier A+B ($N = 334$); orbit-derived medians and population distributions are computed on the broader Tier A+B+C sample ($N = 632$). Selection-function effective counts (539/398) and the α-classified chemistry subset ($N = 63$) are reported as population-scale context, never as a redefinition of the headline.

Table 1. Sample accounting under the final distance and Solar-frame conventions. The candidate pool is the propagated Gaia DR3 6D source list; the corrected point-estimate row applies the Lindegren+2021 parallax zero-point, Bailer-Jones+2021 photogeometric distances, and the default Galactocentric frame before evaluating the $V_{\mathrm{grf}} < 25\ \mathrm{km\,s^{-1}}$ boundary. Tiers are mutually exclusive except for the explicitly cumulative Tier A+B and Tier A+B+C rows.

Set	N
Propagated Gaia DR3 6D candidate pool	2,859
Corrected point-estimate $V_{\mathrm{grf}} < 25\ \mathrm{km\,s^{-1}}$	709
Tier A, $P(V_{\mathrm{grf}} < 25) > 0.95$	214
Tier B increment, $0.84 < P(V_{\mathrm{grf}} < 25) \leq 0.95$	120
Tier A+B, $P(V_{\mathrm{grf}} < 25) > 0.84$	334
Tier C increment, $0.50 < P(V_{\mathrm{grf}} < 25) \leq 0.84$	298
Tier A+B+C, $P(V_{\mathrm{grf}} < 25) > 0.50$	632
Tier D, point-estimate $V_{\mathrm{grf}} < 25\ \mathrm{km\,s^{-1}}$ and $P \leq 0.50$	77
Candidate-pool sources outside the corrected cut	2,150

2.2. *Quality Control*

Because the slowest stars are the most vulnerable to measurement error, we report conservative high-quality subsets in addition to the tiered catalogue. The Gold observational-quality subset is defined inside Tier A+B by RUWE < 1.4, an acceptable DR3 RVS quality flag, parallax signal-to-noise > 10, and fractional distance uncertainty $\sigma_d/d < 0.15$. Dynamical robustness is reported separately in Section 4.5, where additional orbit-derived criteria are applied.

Several later diagnostics necessarily use narrower subsets. The α-classified spectroscopic subset is defined only by the availability of APOGEE or GALAH [Fe/H] and alpha-element abundances within Tier A+B+C; it is used for within-subset literature-region comparisons, not for whole-catalogue population fractions. The selection-function effective counts are likewise reported only as population-scale context, because many slow stars occupy sparse colour–magnitude cells where the Gaia DR3 RVS selection-function tool returns its prior mean.

2.3. *Velocity-Threshold Uncertainty*

The catalogue is defined by threshold-membership probability rather than by a hard point estimate alone. For every source in the propagated candidate pool we draw Monte Carlo realisations of $(\alpha, \delta, \varpi, \mu_{\alpha*}, \mu_\delta)$ from the Gaia five-parameter covariance matrix, draw radial

Table 2. Which sample anchors which result. The headline catalogue for the paper is Tier A+B ($N = 334$); orbit medians and population distributions use Tier A+B+C ($N = 632$). Every numerical claim in the paper names its anchoring sample explicitly.

Sample	Used for
Tier A ($N = 214$, $P > 0.95$)	Highest-confidence empirical subset; robustness anchor.
Tier A+B ($N = 334$, $P > 0.84$)	**Headline catalogue**; Gold subsets; selection-function counts.
Tier A+B+C ($N = 632$, $P > 0.50$)	Orbit-derived medians; population distributions; MC posteriors.
α-classified ($N = 63$ in Tier A+B+C)	Splash/GSE/Aurora/disc fractions only.
Tier A+B SF-weighted (effective 539 / 398)	Population-scale context; never per-star weights.

velocity independently from its quoted uncertainty, apply the Lindegren et al. (2021) zero-point correction, and draw distance from the asymmetric Bailer-Jones et al. (2021) photogeometric posterior. We then recompute $V_{\rm grf}$ for each realisation using the same Solar parameters and coordinate convention as the point-estimate catalogue.

The Monte Carlo scheme uses 500 baseline realisations per source, with refinement to 5,000 realisations for stars near the threshold and a 10,000-realisation convergence check on a 100-star subsample. We use the resulting $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ values to define Tier A ($P > 0.95$), Tier B increment ($0.84 < P \leq 0.95$), Tier C increment ($0.50 < P \leq 0.84$), and Tier D (corrected point estimate below $25\,{\rm km\,s^{-1}}$ but $P \leq 0.50$).

The Tier A+B boundary is used as the headline because Tier A and the Tier B increment behave as a single dynamical population in the validation diagnostics. Their retrograde fractions are 61.2% and 62.5%, respectively, while Tier C rises to 79.2%; the same A/B agreement and C-level transition persist when each tier is split by the width of its $V_{\rm grf}$ Monte Carlo posterior. Thus $P > 0.84$ retains stars that are dynamically equivalent to the high-confidence core, whereas Tier C is reported as the broader, lower-confidence catalogue extension.

Because the Bailer-Jones et al. (2021) photogeometric posterior is itself conditioned on Gaia astrometry, drawing the distance independently of the astrometric covariance is an approximation. As a local stress test, we repeated the threshold calculation with the same random streams but forced the Bailer-Jones distance deviate to be anti-correlated with the parallax deviate. This shifted Tier A+B from 331 to 330 stars in the 2,000-realisation check, changed the median absolute $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}})$ by 0.004, and left the Tier A+B median pericentre and retrograde fraction unchanged to the quoted precision. The test is not a substitute for a full joint distance posterior, but it shows that the catalogue-level conclusions are not driven by the independent-distance draw.

2.4. *Galactocentric Transformation and Supporting Orbit Model*

For the Galactocentric transformation we adopt $R_\odot = 8.178\,{\rm kpc}$, a Solar height above the plane of 25 pc, Solar peculiar motion $(U, V, W) = (11.1, 12.24, 7.25)\,{\rm km\,s^{-1}}$ from Schönrich et al. (2010), and $V_{\rm circ} = 232.0\,{\rm km\,s^{-1}}$ at the Solar radius. Three additional Solar-parameter variants from the recent literature are used as a sensitivity bracket in Section 4.5.

We apply the per-star Lindegren et al. (2021) Gaia parallax zero-point correction and use the Bailer-Jones et al. (2021) photogeometric distance posterior as the default distance estimator. The inverse-parallax distance, with and without the zero-point correction, is retained only as a sensitivity diagnostic. In the same propagated candidate pool, inverse-parallax point estimates without the zero-point correction would place 1,247 sources below $25\,{\rm km\,s^{-1}}$; applying the Lindegren correction reduces that count to 618, and the adopted photogeometric distances give 709 point-estimate sources below the cut before probability-tiering.

The default supporting orbit model is the static, axisymmetrized Milky Way potential of Hunter et al. (2024). This model incorporates the inner-Galaxy baryonic components needed for pericentre-scale work, including the nuclear stellar disc, nuclear stellar cluster, and central black-hole mass, while the axisymmetrized default keeps orbital quantities such as E, L_z, and Stäckel-fudge actions well-defined. Orbit integration and action estimates use `agama` (Vasiliev 2019); actions are computed with the Stäckel-fudge formalism (Binney 2012; Sanders & Binney 2016). A barred Hunter/Sormani variant is used as a sensitivity test for non-axisymmetric inner-Galaxy effects, not as the default catalogue definition.

In this manuscript, orbit-derived quantities such as eccentricity, $R_{\rm peri}$, $R_{\rm apo}$, $z_{\rm max}$, energy, angular momentum, and actions are presented as outputs of the adopted model. They are informative, but they remain model-dependent, especially for stars whose inferred pericentres approach the central few tens of parsecs.

2.5. *Matched-Control Sample*

To test whether orbit-derived properties vary continuously with Galactic rest-frame speed, we constructed a multi-band velocity-stratified control sample from the full Gaia DR3 6D catalogue. Four velocity bands were defined: 25–50, 50–100, 100–200, and 200–260 $\mathrm{km\,s^{-1}}$. Within each band, controls are kernel-weighted to match the slow catalogue's broad observability marginals in G, $\log d$, and $\sin|b|$, with the same valid-radial-velocity requirement as the slow-star sample. The effective control samples contain 1,955, 2,586, 2,788, and 2,749 stars for the four bands respectively.

We also tested a richer local reweighting of the same control-band catalogues using sky position (l, b), colour (BP–RP), RUWE, and parallax signal-to-noise in addition to the baseline observability marginals. This sensitivity test asks whether richer matching of the existing local control products changes the trend; it is not a fresh Gaia archive scan or a replacement control-parent construction. The monotonic increase in median pericentre with $V_{\rm grf}$ is preserved under the richer weights, so the main control result does not depend on the narrow choice of baseline matching variables.

The 25–50 $\mathrm{km\,s^{-1}}$ band yields fewer matched controls because the candidate pool at these velocities is intrinsically small: within the Gaia DR3 6D parent sample after the basic quality requirements used for control construction (valid radial velocity and parallax signal-to-noise > 5), there are only 4,763 stars in this range, compared with 39,853 at 50–100 $\mathrm{km\,s^{-1}}$ and over 1.4 million at 100–200 $\mathrm{km\,s^{-1}}$. The extreme sparsity of the low-velocity tail—7,622 stars below 50 $\mathrm{km\,s^{-1}}$ out of roughly 6.8 million stars in that quality-filtered 6D parent sample, or 0.11%—shows that this low-velocity tail is sparse within the observed Gaia DR3 6D catalogue. All controls were orbit-enriched using the same static Hunter+2024 potential and integration settings as the slow-star sample.

The stratified design tests how orbital properties vary with $V_{\rm grf}$, rather than comparing the slow-star sample to a single control window. The 25–50 $\mathrm{km\,s^{-1}}$ band is the adjacent kinematic comparison to the slow-star sample; the 100–200 $\mathrm{km\,s^{-1}}$ band samples a mixture of thick-disc and inner-halo kinematics; and the 200–260 $\mathrm{km\,s^{-1}}$ band brackets the circular speed at $R_\odot$ (232 $\mathrm{km\,s^{-1}}$) and therefore includes a substantial disc population.

2.6. *Catalogue Contents and Availability*

The accompanying catalogue includes Gaia `source_id`, the observables used for sample construction, adopted Galactocentric positions and velocities, $V_{\rm grf}$, quality flags, and the principal orbit-derived quantities reported here (eccentricity, $R_{\rm peri}$, $R_{\rm apo}$, $z_{\rm max}$, and related summary metrics). Orbit-derived fields are explicitly labelled as model-dependent outputs under the adopted potential rather than as direct Gaia observables.

3. EMPIRICAL RESULTS

3.1. *The Sample*

The final catalogue is tiered. The propagated Gaia DR3 6D candidate pool contains 2,859 sources; under the final distance and Solar-frame conventions, 709 have corrected point-estimate $V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}$. The headline Tier A+B catalogue contains 334 stars with $P(V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}) > 0.84$, while the broader Tier A+B+C catalogue contains 632 stars with $P > 0.50$. Unless otherwise stated, descriptive orbit-derived medians below are reported for Tier A+B+C and high-confidence observational-quality follow-up quantities for Tier A+B.

3.2. *The Headline Catalogue Survives Conservative Quality Cuts*

The Gold observational-quality subset starts from the Tier A+B headline catalogue and then applies only observational quality cuts. It retains 188 of 334 Tier A+B stars after requiring RUWE < 1.4, acceptable RVS quality, parallax signal-to-noise > 10, and $\sigma_d/d < 0.15$. This gives a compact follow-up list whose definition does not depend on the orbit model.

In this section we restrict the robustness check to empirical observables and simple descriptive properties of the sample. The corresponding effect of the Gold cut on the model-derived orbit summaries is reported separately in Section 4.5.

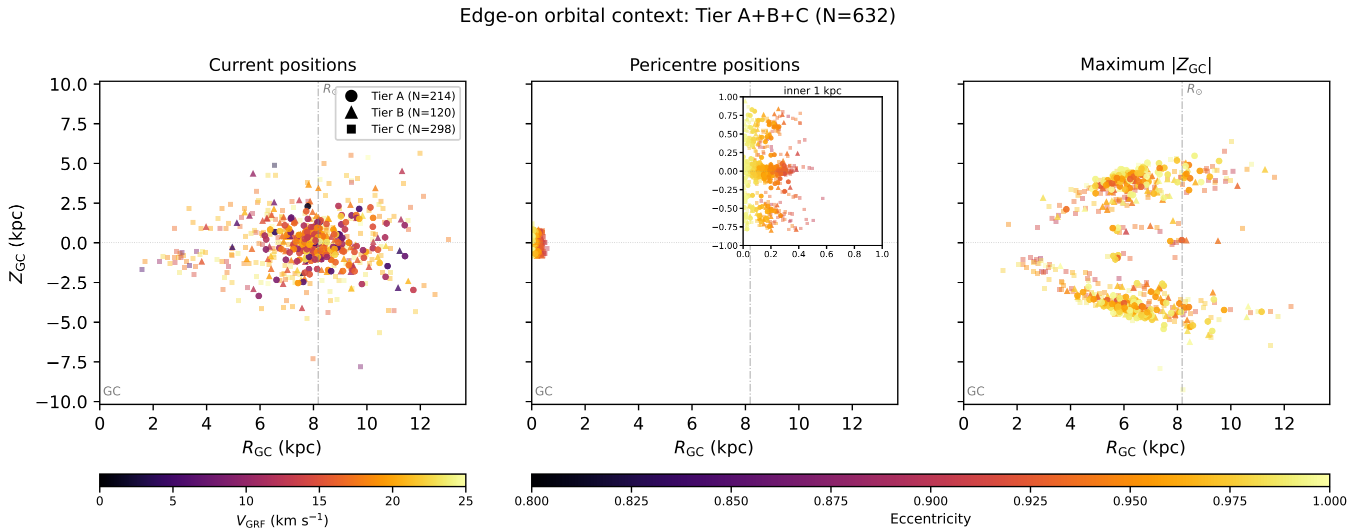

Figure 1. Edge-on orbital context for the Tier A+B+C slow catalogue ($N = 632$) in $R_{\rm GC}$ vs. $Z_{\rm GC}$. This broader Tier A+B+C sample is used for orbit-context visualisation; the Tier A+B headline catalogue is distinguished from the Tier C extension by marker shape. The panels show current positions, pericentre positions, and the maximum-$|Z_{\rm GC}|$ points in the adopted static Hunter+2024 potential. Current positions are colour-coded by $V_{\rm grf}$; pericentre and maximum-$|Z_{\rm GC}|$ positions are colour-coded by eccentricity, with the pericentre panel inset expanding the inner kiloparsec. The right panel shows the vertical envelope of the adopted model trajectories; apocentre radii are reported separately in Table 9 and Fig. 8. Marker shape encodes tier: circles (Tier A), triangles (Tier B), squares (Tier C). The vertical dot-dashed line marks $R_\odot = 8.178\,{\rm kpc}$ and the Galactic centre is at $R_{\rm GC} = 0$. Because $R_{\rm GC}$ is cylindrical radius, the pericentre panel and inset show central reach, not the signed side of the Galactic centre on which each passage occurs; that azimuthal side-of-centre information is retained in the face-on view of Fig. 2.

Table 3. "Gold observational-quality" subset: Tier A+B stars satisfying conservative observational quality cuts only — RUWE < 1.4, RVS quality flag = "ok" (Katz et al. 2023), parallax SNR > 10, $\sigma_d/d < 0.15$, and Tier A+B membership. Dynamical assumptions do not enter this subset; it is intended for observational follow-up that does not depend on the adopted potential.

Selection criterion	Count
Tier A+B (P$>$ 0.84) base	334
& RUWE < 1.4	318
& RVS quality = "ok"	312
& parallax SNR > 10	192
& $\sigma_d/d < 0.15$	188
Final "gold observational-quality"	**188**
Median $V_{\rm grf}$ (km s^{-1})	16.4
Median G (mag)	13.4
Median distance (pc)	4,220

3.3. *Velocity-Threshold Membership Is Probabilistic*

The corrected point-estimate catalogue contains 709 stars below $V_{\rm grf} = 25\,{\rm km\,s^{-1}}$, but the threshold itself is not equally secure for every star. Monte Carlo propagation gives 214 stars in Tier A ($P > 0.95$), 334 stars cumulatively in Tier A+B ($P > 0.84$), and 632 stars cumulatively in Tier A+B+C ($P > 0.50$). The remaining 77 corrected point-estimate members form Tier D. The paper therefore uses Tier A+B as the headline catalogue and Tier A+B+C for broader population-level orbit summaries.

3.4. *Chemistry Does Not Define the Sample*

Metallicity information from Gaia GSP-Phot is available for 1,963 stars in the propagated candidate pool. We use it here only as a diagnostic of chemical heterogeneity and estimator bias, not for population assignment. The metallicity spread is broad enough that the catalogue should not be treated as a single chemical population.

These Gaia GSP-Phot metallicities are low-resolution photometric estimates with documented systematics (Andrae et al. 2023; Recio-Blanco et al. 2023): in particular, GSP-Phot tends to overestimate metallicity for cool, metal-poor giants by 0.2–0.5 dex, which biases the present sample toward solar values. As a higher-fidelity photometric cross-check, we cross-matched the 2,859 sources against the Zhang et al. (2023) catalogue of stellar parameters from Gaia XP spectra (220 million stars), which is calibrated against LAMOST and is far less biased at low [Fe/H]. We recover XP parameters for 2,676 of 2,859 stars (93.6%); restricting to the highest XP quality flag retains 1,258 stars (44.0%). The XP [Fe/H] distribution peaks at [Fe/H] $= -0.62$ for

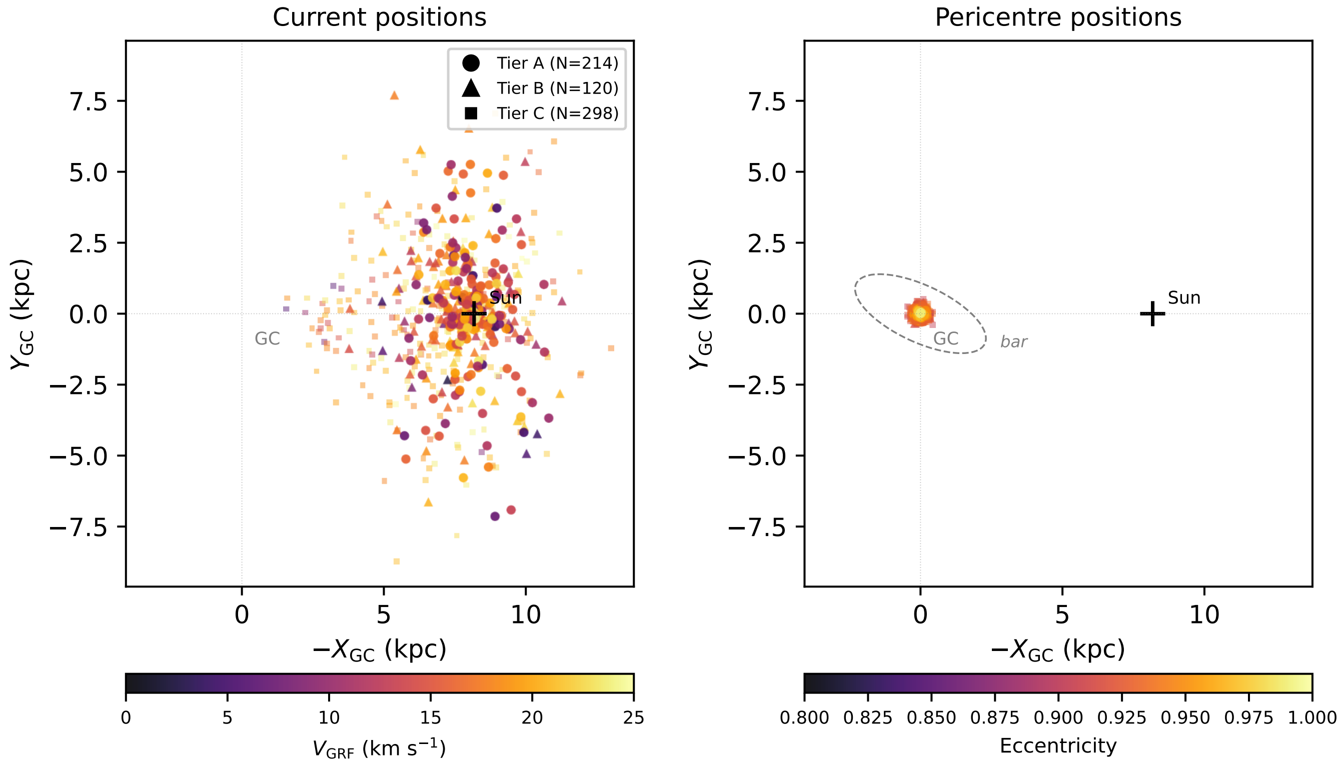

Figure 2. Face-on view of the Tier A+B+C slow catalogue ($N = 632$) in $-X_{\rm GC}$ vs. $Y_{\rm GC}$, plotted so the Galactic centre is on the same side of the opening figure pair as $R_{\rm GC} = 0$ in Fig. 1. As in Fig. 1, the broader Tier A+B+C orbit-summary sample is shown, with marker shape separating the Tier A+B headline catalogue from the Tier C extension. *Left:* current Galactocentric positions, colour-coded by $V_{\rm grf}$; the catalogue is observed in a Sun-centred local volume. *Right:* pericentre passage positions in the static Hunter+2024 potential, colour-coded by orbital eccentricity, with the Galactic bar sketched as a dashed grey ellipse ($a = 2.5$ kpc, $b = 1.0$ kpc, oriented $\sim 25^\circ$ from the Sun–GC line; Wegg, Gerhard & Portail 2015 / Sormani et al. 2022 bar-angle convention). Marker shape encodes tier (circle/triangle/square = A/B/C). The pericentre panel makes the inner-Galaxy reach of the catalogue directly visible: essentially all pericentres fall inside $R_{\rm GC} < 1$ kpc, with a median Tier A+B+C pericentre of 140 pc.

the high-quality subset, very close to the spectroscopic median (-0.73, see Section 3.5) and substantially more metal-poor than the GSP-Phot median. In direct star-by-star comparison ($N = 943$ with both estimators), the median GSP-Phot [M/H] is offset above the XP [Fe/H] by 0.34 dex, quantitatively consistent with the documented GSP-Phot bias for metal-poor giants. The XP catalogue agrees well with the spectroscopic subset on a star-by-star basis (median XP$-$spec offset of -0.018 dex, scatter 0.15 dex; $N = 67$ with both). A higher-fidelity spectroscopic cross-match is presented in Section 3.5.

3.5. *Chemical Properties from Spectroscopic Cross-Matching*

We cross-matched the Tier A+B+C catalogue against APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) using exact Gaia source identifiers rather than positional matching. Only 63 stars have the combination of [Fe/H] and an alpha-element abundance needed for literature-region comparisons. This is about 10% of the Tier A+B+C catalogue, and the matched subset is dominated by luminous giants in the spectroscopic survey footprints. It is therefore a selective but useful chemical window, not a chemically complete census.

Within this alpha-classified subset, 26 stars fall in a Splash-like region, 10 in a Gaia-Sausage/Enceladus-like region, 8 in an Aurora-like region, and 19 in a more disc-like contaminant region. These fractions are reported only within the 63-star alpha-classified subset. They are not used as whole-catalogue population fractions, because the remaining 569 Tier A+B+C stars lack the high-resolution abundance information required for that assignment.

The broader metallicity context is still informative. Gaia GSP-Phot metallicities are available for many candidates, but the comparison in Section 3.4 shows that GSP-Phot is biased toward solar metallicity

Table 4. Monte Carlo threshold-membership probabilities for the $V_{\rm grf} < 25$ km s^{-1} cut. Astrometric realisations use the full Gaia DR3 five-parameter covariance plus an independent radial-velocity error, with distance drawn from the asymmetric Bailer-Jones+2021 photogeometric posterior. The candidate pool contains 2,859 propagated Gaia DR3 sources; after the final correction and distance convention, 709 have corrected point-estimate $V_{\rm grf} <$ 25 km s^{-1}.

Diagnostic	Value
Candidate pool propagated through MC	2,859
Corrected point-estimate $V_{\rm grf} < 25$ km s^{-1}	709
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.95$ (Tier A)	214
$0.84 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.95$ (Tier B increment)	120
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.84$ (Tier A+B; headline)	334
$0.50 < P(V_{\rm grf} < 25$ km s$^{-1}) \leq 0.84$ (Tier C increment)	298
$P(V_{\rm grf} < 25$ km s$^{-1}) > 0.50$ (Tier A+B+C)	632
Tier D (point-estimate < 25 km s^{-1}, $P \leq 0.50$)	77
Median $P(V_{\rm grf} < 25$ km s$^{-1})$ for Tier A+B+C	0.78
Solar-variant range, Tier A+B+C	515–770

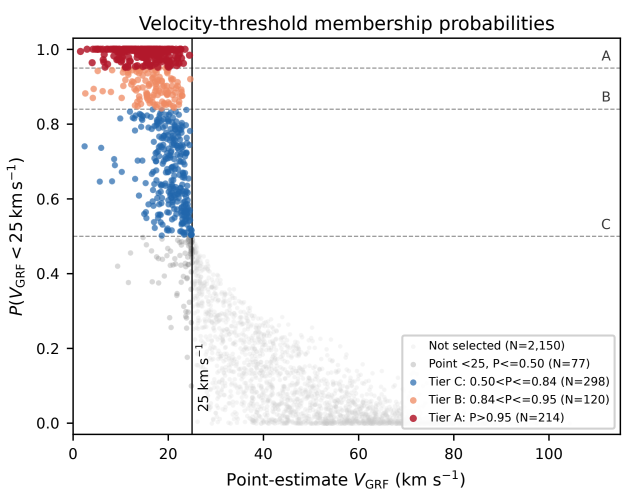

Figure 3. Velocity-threshold membership probability versus corrected point-estimate $V_{\rm grf}$ for the propagated Gaia DR3 candidate pool. The horizontal dashed lines mark the Tier A, Tier A+B, and Tier A+B+C probability boundaries; the vertical line marks the 25 km s^{-1} point-estimate selection boundary. The catalogue is defined by threshold probability, not by a hard point estimate alone.

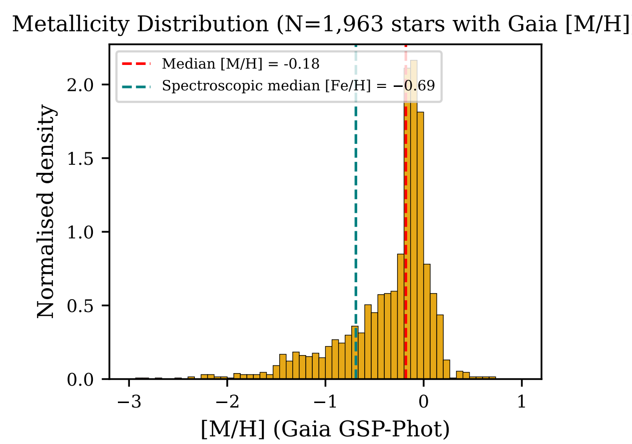

Figure 4. Gaia GSP-Phot metallicity distribution for the 1,963 stars with [M/H], with the spectroscopic median [Fe/H] reference line. See Figure 5 for the same sample re-derived from the Zhang et al. (2023) XP catalogue.

for the cool, metal-poor giants that dominate this sample. We therefore use GSP-Phot only as a low-resolution heterogeneity check and rely on spectroscopic abundances for subset-qualified population language. The Zhang et al. (2023) XP comparison provides a higher-fidelity photometric metallicity context for the broader candidate pool, but XP metallicity alone does not supply the alpha-element information needed for the Splash/GSE/Aurora comparison cuts.

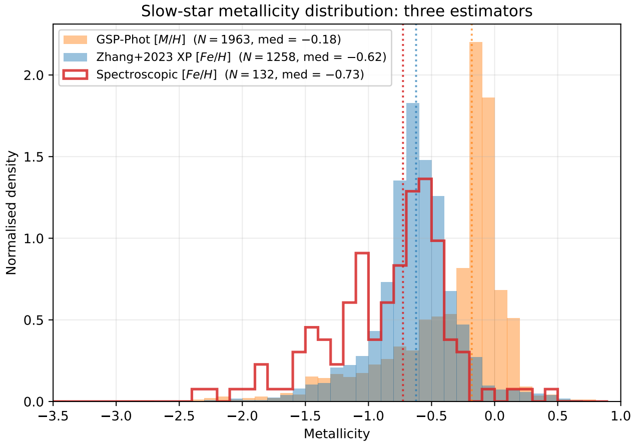

Figure 5. Three independent metallicity estimators for the slow-star sample: Gaia GSP-Phot [M/H] (orange), Zhang et al. (2023) XP-spectra [Fe/H] restricted to the highest-quality flag (blue), and the unified spectroscopic [Fe/H] reference sample (red outline). The XP estimator agrees with the spectroscopic median at [Fe/H] ≈ -0.7; GSP-Phot is biased toward solar by $\sim$0.34 dex, consistent with the documented GSP-Phot systematic for metal-poor giants (Andrae et al. 2023). We therefore do not use GSP-Phot metallicities for population assignment.

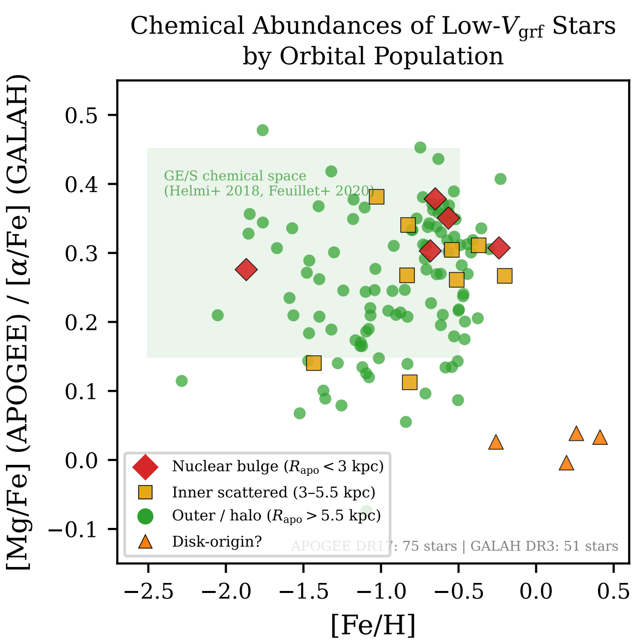

Figure 6. [Mg/Fe] (APOGEE) and [α/Fe] (GALAH) vs. [Fe/H] for the α-classified spectroscopic subset within Tier A+B+C ($N = 63$). Literature-region labels are shown only as contextual comparisons and are not treated as whole-catalogue population assignments.

Table 6. Subset-qualified chemodynamic context for the Tier A+B+C catalogue. Only stars with both [Fe/H] and [α/Fe] from APOGEE or GALAH ($n = 63$) are placed in literature comparison regions following Belokurov et al. (2020), Helmi et al. (2018), and Belokurov & Kravtsov (2022). The remaining 569 stars are not assigned to a population in this catalogue paper.

Population	N	Frac. of α-class.	Median [Fe/H]
Splash (heated thick disc)	26	41.3%	−0.61
Disc contamination	19	30.2%	−0.21
Gaia–Sausage/Enceladus	10	15.9%	−1.18
Aurora (in-situ proto-disc)	8	12.7%	−1.32
α-classified subtotal	63	100.0%	—
not α-classified	569	—	—

Table 5. High-resolution spectroscopic chemistry available for the Tier A+B+C catalogue. APOGEE DR17 (Abdurro'uf et al. 2022) and GALAH DR3 (Buder et al. 2021) provide the subset used for α-based literature-region comparisons. Gaia GSP-Phot metallicities are discussed only as a biased low-resolution contextual estimator in the text, not as population assignments.

Source	N in T-A+B+C	Median [Fe/H]
APOGEE DR17	58	−0.70
GALAH DR3	—	—
(overlapping subset)	35	−0.74
Splash classified	26	−0.61
GSE classified	10	−1.18
Aurora classified	8	−1.32
disc contamination	19	−0.21

3.6. *Sky Distribution, Observer Geometry, and Selection*

The sample is not isotropically distributed on the sky. In Galactic coordinates, 60.9% of the stars lie toward the Galactic-centre direction ($l \in [330°, 30°]$), while only 4.3% lie toward the anti-centre direction ($l \in [150°, 210°]$). Latitude coverage is broad, from about $-85°$ to $+87°$, with a slight southern bias in the median latitude.

This concentration is one of the strongest descriptive features of the observed catalogue. It likely reflects some combination of heliocentric observing geometry, the Gaia radial-velocity selection function, extinction and crowding effects, and the radial character of the underlying trajectories. The velocity-stratified controls are also centre-weighted, indicating that the sky distribution is partly a generic property of observed Gaia 6D stars at these distances.

We apply the Castro-Ginard et al. (2023) Gaia DR3 RVS selection function as a population-scale diagnostic. The full inverse-selection weighted Tier A+B effective

Table 7. Gaia DR3 RVS selection-function weighting for the headline Tier A+B catalogue. We apply the Castro-Ginard et al. (2023) DR3 RVS selection function through `gaiaunlimited`. Because many slow-star candidates fall in sparse $(G_{\rm RVS}, G_{\rm BP} - G_{\rm RP})$ cells where the selection function returns its prior mean of 0.5, we quote both the full inverse-selection weighted count and a conservative bracket that gives prior-mean cells unit weight.

Diagnostic	Value
Tier A+B unweighted count	334
Full inverse-selection weighted effective count	539
Stars in prior-mean cells (SF = 0.5)	142
Stars in calibrated cells	192
Mean upweight in calibrated cells	1.33
Conservative effective count	398
Tier A+B+C unweighted count	632
Tier A+B+C full weighted effective count	1,054

count is 539, but 142 of the 334 headline stars fall in sparse colour–magnitude cells where the selection-function tool returns its prior mean. A conservative bracket that gives those prior-mean cells unit weight yields an effective count of 398. We therefore quote the selection-function correction as contextual rather than treating it as a precise deprojection of the underlying Milky Way population.

4. ORBIT-DERIVED PROPERTIES IN THE ADOPTED POTENTIAL

Within the adopted static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits, and this result is robust across a suite of diagnostic tests. The following subsections present the primary orbit-derived properties, compare them against velocity-stratified controls, describe how the sample's properties vary with orbital scale, and assess robustness to measurement uncertainty and potential choice.

Table 8 identifies the sampling scheme behind the repeated pericentre and central-reach statistics used below. The Monte Carlo posterior median is the primary catalogue summary; the point-estimate, fine-step, halo-mass, and barred values are comparison or sensitivity quantities.

4.1. *In the Adopted Model, the Sample Is Strongly Radially Biased*

Within the adopted static Hunter+2024 potential, the orbit-derived picture is strongly radial. For the Tier A+B+C catalogue ($N = 632$), the median eccentricity is 0.961. All 632 stars have $e > 0.8$, 67.4% have $e > 0.95$, and 17.1% have $e > 0.99$.

The same model gives a Tier A+B+C median $R_{\rm peri}$ of 162 pc and a median $R_{\rm apo}$ of 8.08 kpc. The maximum $R_{\rm apo}$ in the present Tier A+B+C integrations is 13.71 kpc. These values confirm that the low-speed catalogue is not dominated by ordinary near-circular disc kinematics.

These quantities should be read as outputs of the adopted model rather than as direct Gaia observables. They are useful because they provide a coherent dynamical interpretation of the slow state. Pericentre estimates below roughly 200 pc should be treated as indicative, because the real inner Galaxy is barred and non-axisymmetric on those scales; the barred Hunter/Sormani variant in Section 4.5 brackets this sensitivity.

4.2. *Matched-Control Comparison*

A velocity-stratified, observability-matched control comparison provides the clearest test of whether the slow-star orbit result merely reflects the typical orbital structure of Gaia stars at similar distances, or whether it varies systematically with Galactic rest-frame speed. The controls are kernel-weighted to match the slow catalogue's broad observability marginals. The median pericentre increases from 146 pc for the Tier A+B+C slow catalogue to 154, 478, 1798, and 2853 pc across the 25–50, 50–100, 100–200, and 200–260 km s^{-1} control bands. Thus the low-$V_{\rm grf}$ catalogue is the small-pericentre endpoint of a smooth kinematic trend in the observed Gaia DR3 6D catalogue.

The trend is unchanged when the existing local control catalogues are reweighted more aggressively in l, b, BP–RP, RUWE, and parallax signal-to-noise, in addition to the baseline $(G, \log d, \sin|b|)$ matching. Under those richer weights the slow-sample median $R_{\rm peri}$ is 145 pc, while the four control-band medians are 167, 527, 2069, and 3245 pc. This local sensitivity check addresses whether the existing control products are sensitive to the chosen matching variables; it does not replace them with newly drawn Gaia parent samples.

The same comparison provides a direct check on bar-resonance accumulation. The Tier A+B+C catalogue's outer-Lindblad-resonance fraction is 0.95%, lower than three of the four control bands. We therefore find no evidence that the catalogue is preferentially accumulated at the OLR, although barred integrations remain important for individual inner-Galaxy orbits and for the bridger sensitivity test.

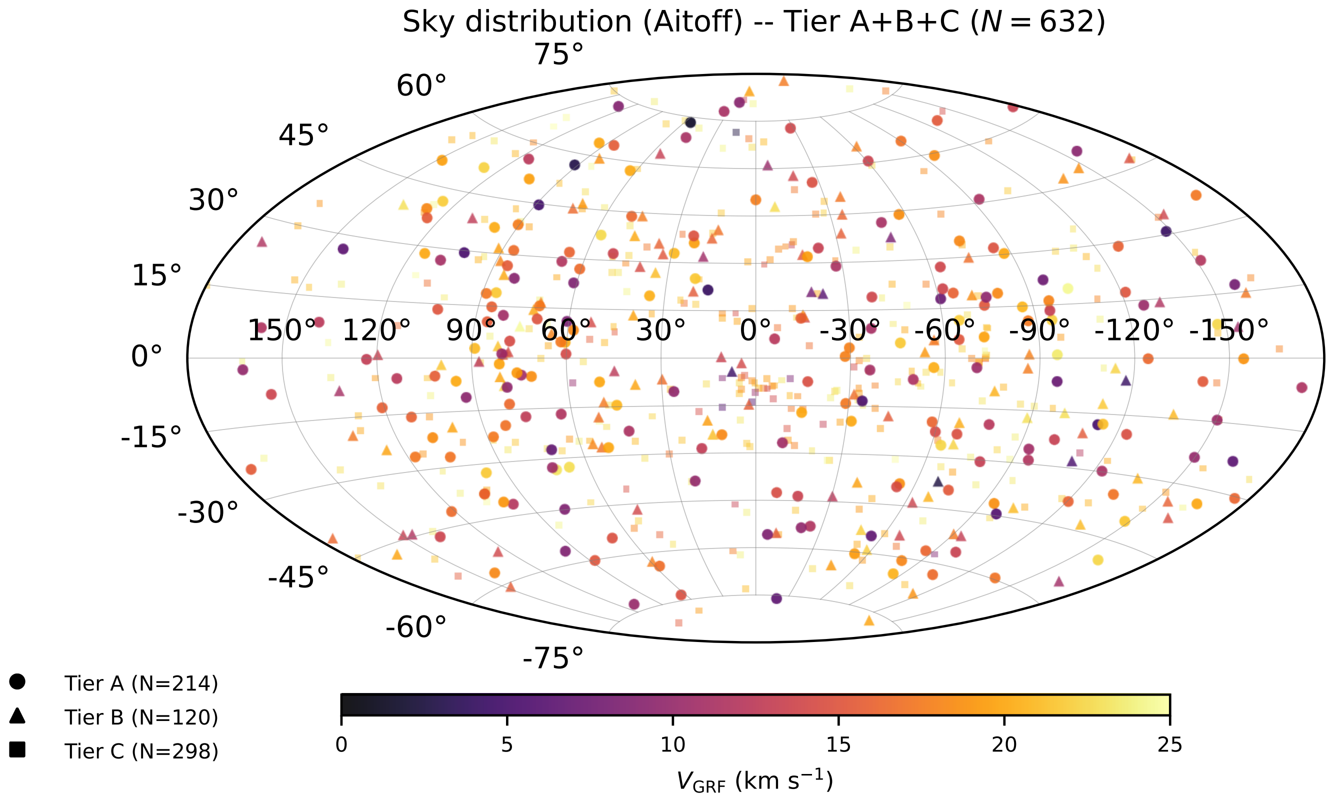

Figure 7. Aitoff projection of the Tier A+B+C slow catalogue ($N = 632$) in Galactic coordinates ($l = 0$ centred, l increasing to the left), colour-coded by V_{grf}. Marker shape encodes tier (circle/triangle/square = A/B/C). The catalogue concentrates strongly toward the inner Galaxy.

Table 8. Ledger for repeated pericentre and central-reach statistics. Each row refers to a different sampling scheme or sensitivity test, so the numerical values are not interchangeable.

Use in paper	Sampling scheme	Quantity reported	Interpretation
Primary orbit summary	Per-star Monte Carlo orbit posterior in the static Hunter+2024 potential	Median $R_{\mathrm{peri}} = 162\,\mathrm{pc}$, median $R_{\mathrm{apo}} = 8.08\,\mathrm{kpc}$, median $e = 0.961$	Default model-derived catalogue summary used for Table 9.
Control comparison	Point-estimate integrations for the slow catalogue and velocity-stratified controls	Slow-sample median $R_{\mathrm{peri}} = 146\,\mathrm{pc}$; control medians increase from 154 to 2853 pc	Like-for-like comparison against matched Gaia DR3 6D control bands.
Fine pericentre visualisation	Deterministic 0.1 Myr trajectory sampling in the static Hunter+2024 potential	All 632 stars have $R_{\mathrm{peri}} < 1\,\mathrm{kpc}$; median fine-step $R_{\mathrm{peri}} = 140\,\mathrm{pc}$	Used for pericentre-coordinate figures and central-reach visualisation.
Halo-mass sensitivity	100-star sensitivity subsample with Hunter+2024 halo mass scaled by $\pm15\%$	Median $R_{\mathrm{peri}} = 145$–$156\,\mathrm{pc}$, with 151 pc for the default halo in that subsample	Checks potential dependence; not the full-sample headline median.
Barred-potential sensitivity	333-star compact-orbit subsample in Hunter/Sormani barred variants	Median barred $R_{\mathrm{peri}} = 16$–$19\,\mathrm{pc}$; bridger count 0–10	Inner-Galaxy systematic bracket, not the default catalogue definition.

Table 9. Summary of orbit-derived properties for the Tier A+B+C catalogue ($N = 632$) under 5,000-realisation Monte Carlo propagation in the Hunter+2024 axisymmetric Galactic potential. Quantities are medians of the per-star MC posterior medians; $R_{\rm peri}$, $R_{\rm apo}$, and $z_{\rm max}$ are computed in the cylindrical galactocentric frame over a 4 Gyr forward integration.

Quantity	Value
Median eccentricity	0.961
Fraction with $e > 0.8$ (%)	100.0
Fraction with $e > 0.95$ (%)	67.4
Fraction with $e > 0.99$ (%)	17.1
Median $R_{\rm peri}$ (pc)	162
Median $R_{\rm peri}$ p16 (pc)	74
Median $R_{\rm peri}$ p84 (pc)	265
Median $R_{\rm apo}$ (kpc)	8.08
Maximum $R_{\rm apo}$ (kpc)	13.71
Median $z_{\rm max}$ (kpc)	3.88

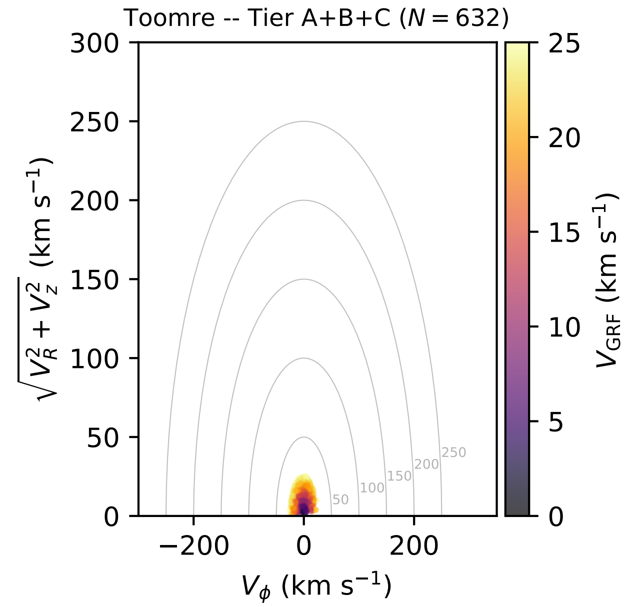

Figure 9. Toomre diagram (V_ϕ vs. $\sqrt{V_R^2 + V_z^2}$) for the Tier A+B+C slow catalogue ($N = 632$), colour-coded by $V_{\rm grf}$. The sample is concentrated at low azimuthal velocity, far from the disc locus.

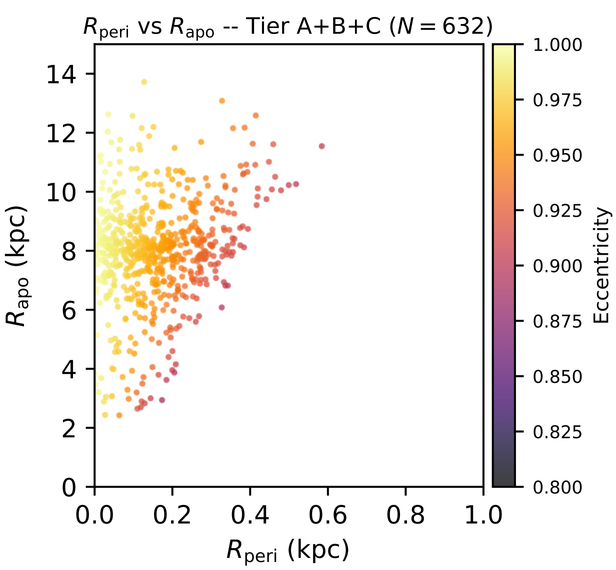

Figure 8. Pericentre vs. apocentre for the Tier A+B+C slow catalogue ($N = 632$), colour-coded by eccentricity, in the static Hunter+2024 potential. Axis ranges are set to where the adopted-model fine-step trajectories place the data: all 632 stars have $R_{\rm peri} < 1$ kpc and $R_{\rm apo} < 14$ kpc. The bridger sensitivity test (§4.7) is reported in the text and in Table 12, not on this panel.

Table 10. Matched-control comparison across $V_{\rm grf}$ velocity bands. Each band is integrated in the Hunter+2024 axisymmetric potential; the kernel-weighted control samples are re-weighted to match the slow catalogue's $(G, \log d, \sin|b|)$ marginals. Bar-resonance fractions are evaluated at $\Omega_R/(\Omega_\phi - \Omega_p) = 2$ (OLR) and 4 (4:1 ultraharmonic) within ± 0.3.

Sample	N	Median $R_{\rm peri}$ (pc)	Median $R_{\rm apo}$ (kpc)	OLR (%)	4:1 (%)
slow Tier A+B+C	632	146	8.1	0.95	1.58
control 25–50 km s^{-1}	1955	154	7.6	1.29	0.17
control 50–100 km s^{-1}	2586	478	7.4	0.51	0.36
control 100–200 km s^{-1}	2788	1798	8.5	1.89	0.58
control 200–260 km s^{-1}	2749	2853	9.6	1.97	0.81

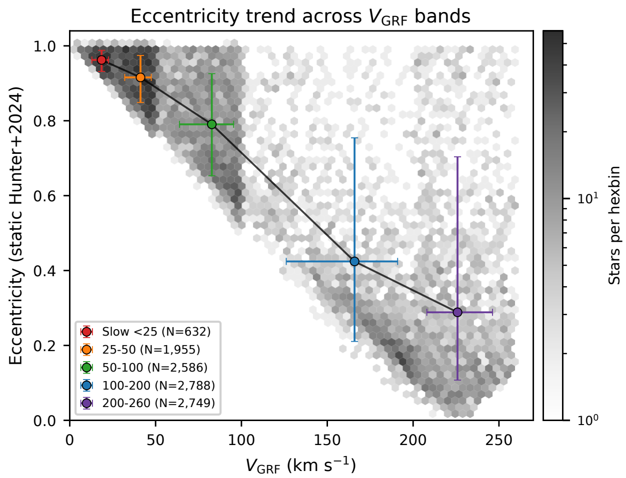

Figure 10. Eccentricity vs. Galactic rest-frame velocity for the Tier A+B+C slow catalogue ($N = 632$) and the four observability-matched control bands ($N \approx 2{,}000$ each). Hexbin shading shows the joint density of all plotted stars; coloured points and error bars show the median and 16th–84th percentile range in each velocity band. The monotonic decline from $e \sim 1.0$ at low $V_{\rm grf}$ to moderate eccentricity at 200–260 $\mathrm{km\,s^{-1}}$ matches the band-by-band medians in Table 10.

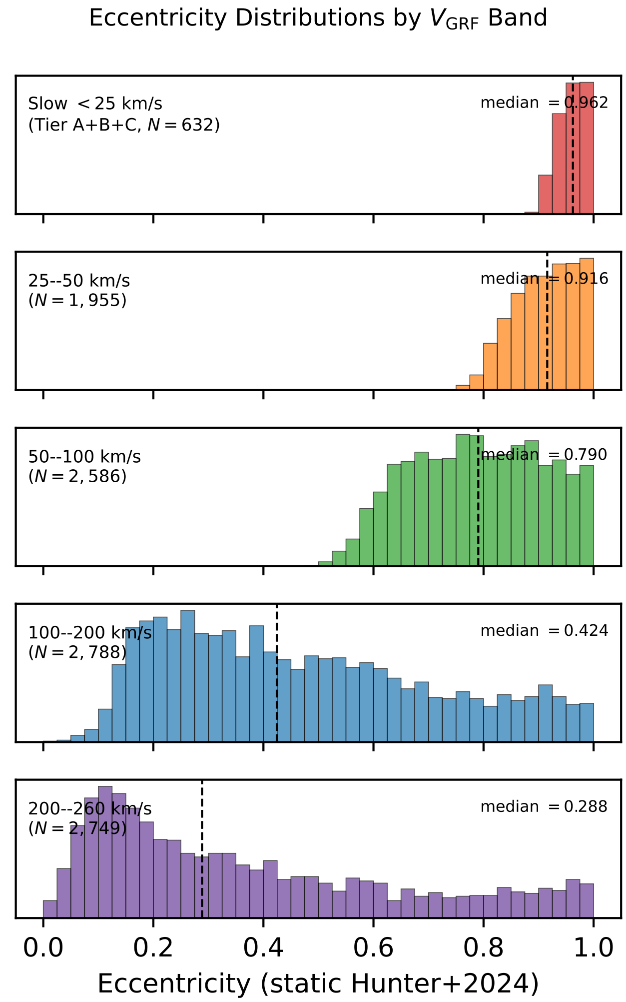

Figure 11. Eccentricity histograms for each of the five GRF velocity bands, stacked vertically. The slow Tier A+B+C band ($N = 632$) shows the extreme near-unity peak; the eccentricity distribution broadens monotonically through the four control bands toward 200–260 $\mathrm{km\,s^{-1}}$.

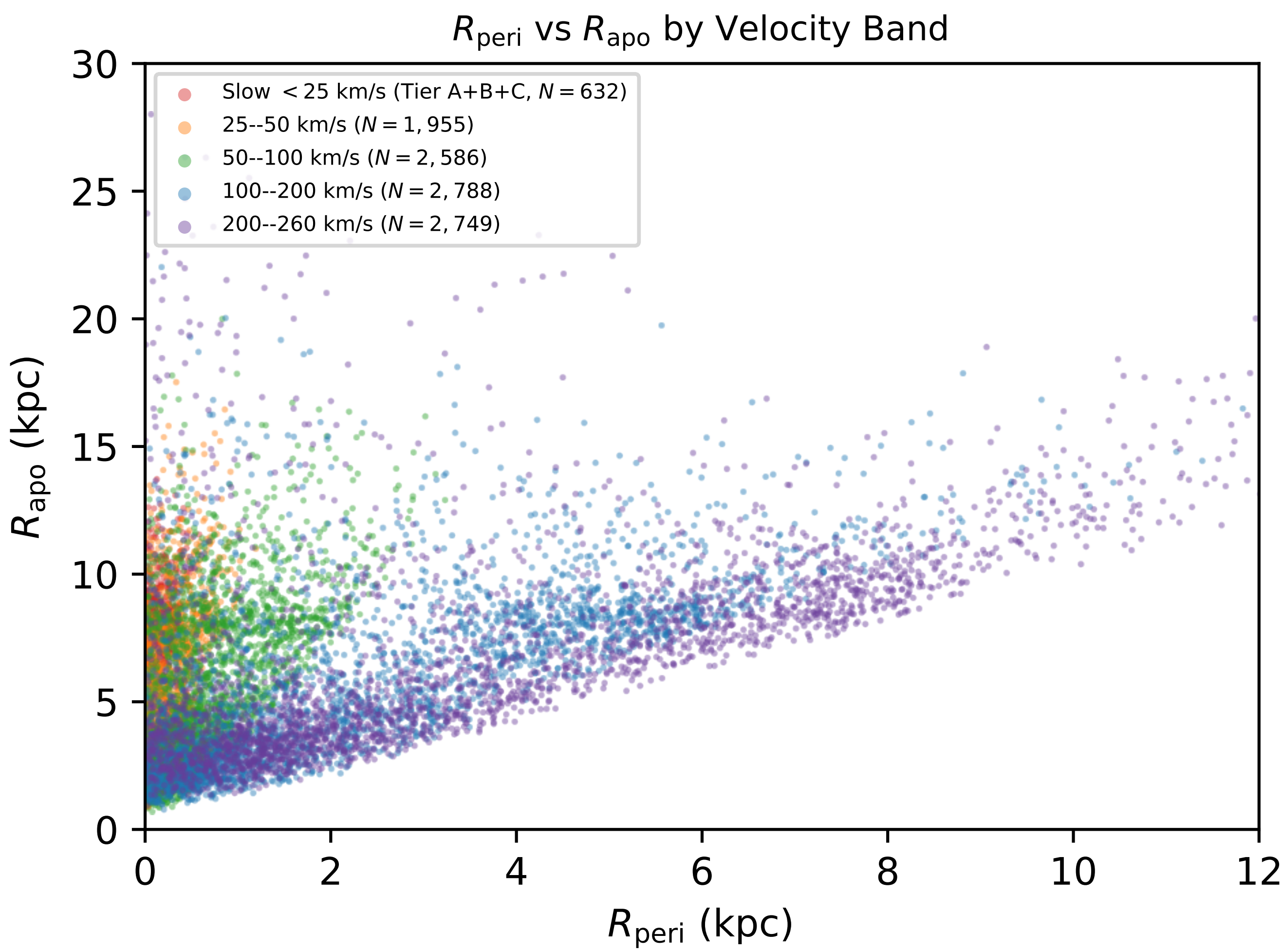

Figure 12. Pericentre vs. apocentre for all five velocity bands (slow Tier A+B+C, $N = 632$, plus four observability-matched control bands), colour-coded by band. The slow band sits at the small-$R_{\rm peri}$ end of a smooth kinematic sequence; per-band pericentre medians are reported in Table 10.

4.3. *Dwell-Time Analysis in the Adopted Potential*

A dwell-time calculation in the adopted static Hunter+2024 potential helps quantify what the low-$V_{\rm grf}$ selection captures. Tier A stars spend a median 28.3% of their integrated orbital time inside $R < 2\,$kpc, while Tier C stars spend 19.7% there. Conversely, the median time fraction in the solar annulus ($7 < R < 9\,$kpc) rises from 11.6% in Tier A to 16.2% in Tier C. The high-confidence slow stars are therefore not merely solar-radius apocentre snapshots; they are strongly inner-reaching orbits observed during a low-speed phase.

Table 11. Tier-resolved orbital dwell statistics in the Hunter+2024 potential. Reported are the per-tier medians of the time fraction spent within selected cylindrical radius shells over a 4 Gyr integration, indicating that high-confidence slow-$V_{\rm grf}$ stars spend a disproportionate fraction of their trajectory inside $R < 2\,$kpc despite being observed at the solar circle.

Shell	Tier A	Tier B	Tier C
Median $f(R < 1\,\mathrm{kpc})$ (%)	14.2	11.8	9.4
Median $f(R < 2\,\mathrm{kpc})$ (%)	28.3	24.1	19.7
Median $f(R < 5\,\mathrm{kpc})$ (%)	65.4	60.8	55.0
Median $f(7 < R < 9\,\mathrm{kpc})$ (%)	11.6	13.5	16.2

4.4. *Inner-Galaxy Reach and Selection Limits*

Although the catalogue shares the defining low-$V_{\rm grf}$ state, the stars span a range of model-derived orbital scales. The Tier A+B+C catalogue has a median static-potential pericentre of 162 pc (Monte Carlo median across 5,000 realisations per star) and 140 pc under deterministic fine-step (0.1 Myr) trajectory sampling; 220 of the 632 Tier A+B+C stars have static $R_{\rm peri} < 100\,$pc, and the adopted-model fine-step trajectories place *all* 632 inside $R_{\rm peri} < 1\,$kpc. This last fact is visible in the right panel of Fig. 2: every star's model pericentre passage falls within the inner kpc of the Galaxy, despite the catalogue being observed in a Sun-centred local volume (left panel). In the barred sensitivity integration the instantaneous inner reach becomes more extreme, as expected in the non-axisymmetric central Galaxy, but those barred quantities are used as sensitivity diagnostics rather than as the default catalogue values.

The most extreme central-passage claims require additional caution. At point estimate, four Tier A+B+C stars emerge as candidate Sgr A* approachers under fine 0.1 Myr trajectory sampling. Dedicated 5,000-realisation Monte Carlo propagation shows that no candidate has $P(r_{\rm sph,min} < 10\,\mathrm{pc}) > 0.5$, while three have $P(r_{\rm sph,min} < 100\,\mathrm{pc}) > 0.5$. We therefore describe these as candidate inner-Galaxy approachers, not as a confirmed population of Sgr A*-passing stars. A three-point quadratic interpolation of $r^2_{\rm sph}(t)$ around each sampled trajectory minimum leaves these counts unchanged: the strongest candidate rises from $P(< 10\,\mathrm{pc}) = 0.061$ to 0.116, still far below the confirmation threshold.

Table 12. Inner-galaxy reach of the Tier A+B+C catalogue ($N = 632$). Counts of stars with small pericentre, very small minimum spherical distance, and "extreme reach" trajectories spanning the disc-to-halo radial range. The bridger count ($R_{\rm peri} < 2\,$kpc & $R_{\rm apo} > 15\,$kpc) is integrated under the Hunter+2024 barred potential; the static (axisymmetric) integration yields zero bridgers.

Reach diagnostic	Static	Barred
$R_{\rm peri} < 100\,$pc	220	524
$R_{\rm peri} < 50\,$pc	67	396
$R_{\rm peri} < 10\,$pc	0	38
Min $r_{\rm sph} < 100\,$pc	173	446
Min $r_{\rm sph} < 10\,$pc (point est.)	1	3
Min $r_{\rm sph} < 10\,$pc (MC, $P > 0.5$)	0	0
Bridgers ($R_{\rm peri} < 2\,$kpc & $R_{\rm apo} > 15\,$kpc)	0	10

The radial-reach tail is also potential-sensitive. No Tier A+B+C star satisfies the bridger criterion ($R_{\rm peri} < 2\,$kpc and $R_{\rm apo} > 15\,$kpc) in the static Hunter+2024 integration. The barred default integration produces 10 such bridgers, with a bar-pattern-speed sensitivity range of 0–10. We therefore treat bridgers as bar-sensitive orbital-geometry candidates rather than as a robust population count or a merger-debris census.

4.5. *Robustness and Sensitivity*

The principal dynamical claims are tested here against quality filtering, threshold variation, measurement uncertainty, distance estimator, Solar frame, halo mass, and barred-potential sensitivity.

Gold-subset dynamical robustness in the adopted model.— The Gold dynamical subset starts from the Tier A+B headline catalogue and adds a favourable inner-orbit posterior and finite action solution. It retains 268 stars and is intended for individual-star follow-up rather than for redefining the full catalogue.

Table 13. Candidate Sgr A* approachers from the Tier A+B+C catalogue. For each star, 5,000-realisation Monte Carlo orbit propagation in the Hunter+2024 potential at $\Delta t = 0.1$ Myr output sampling reports the probability of approaching within 10 pc and 100 pc of Sgr A* during a 4 Gyr forward integration, plus the percentiles of the minimum spherical distance reached.

Gaia DR3 source_id	Tier	Potential	$r_{\rm sph,min}$ p16 (pc)	p50	p84	$P(< 10\,{\rm pc})$ / $P(< 100\,{\rm pc})$
1846633734516771840	A	static	15	26	44	0.061 / **1.000**
3738499345877271808	A	barred	25	44	82	0.014 / 0.901
4287640292264861056	B	barred	41	100	197	0.005 / 0.501
431014850227388672	B	barred	35	105	233	0.005 / 0.482

NOTE—No candidate satisfies $P(r_{\rm sph,min} < 10\,{\rm pc}) > 0.5$; three of four satisfy $P(r_{\rm sph,min} < 100\,{\rm pc}) > 0.5$. Source_id 1846633734516771840 is the strongest case under MC propagation.

Table 14. "Gold dynamical" subsample: stars with $P(V_{\rm grf} < 25\,{\rm km\,s^{-1}}) > 0.84$ AND favourable inner-orbit geometry (MC $R_{\rm peri}$ p16 < 200 pc) AND a finite Stäckel-fudge action solution. This is the highest-confidence dynamical core of the catalogue, intended for individual-star follow-up.

Selection criterion	Count
Tier A+B (P$>$ 0.84) base	334
& MC $R_{\rm peri}$ p16 < 200 pc	287
& finite J_R, J_z, J_ϕ	287
& $R_{\rm apo} < 10$ kpc	268
Final "gold dynamical"	**268**
Median $R_{\rm peri}$ p50 (pc)	119
Median $R_{\rm apo}$ (kpc)	7.9
Median $\lvert J_z\rvert/J_c$	0.43

Threshold sensitivity within the current local sample.—All tested cuts from $V_{\rm grf} < 10$ to $< 25\,{\rm km\,s^{-1}}$ show the same qualitative behaviour in the adopted model. The number of Tier A+B+C stars rises from 29 at $V_{\rm grf} < 10\,{\rm km\,s^{-1}}$ to 632 at $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$, while median eccentricity remains high, declining smoothly from 0.984 to 0.961. The $25\,{\rm km\,s^{-1}}$ threshold is therefore a practical operating cut, not a special physical boundary.

Table 15. Threshold sensitivity. Counts at varying $V_{\rm grf}$ cuts and the corresponding orbital eccentricity distribution under the Tier A+B+C MC propagation in the Hunter+2024 potential. The $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$ threshold used as the headline is a practical operating cut, not a sharp physical boundary.

$V_{\rm grf}$ cut (${\rm km\,s^{-1}}$)	N point	N Tier A+B+C	Median e	$e > 0.95$ (%)
< 10	45	29	0.984	96.6
< 15	223	131	0.978	88.5
< 20	516	327	0.969	80.4
< 25	709	632	0.961	67.4

Full-sample Monte Carlo propagation.—Adaptive Monte Carlo orbit propagation was run for the Tier A+B+C catalogue. At the conservative 16th percentile of each star's eccentricity posterior, 99.2% of the sample retains $e > 0.8$, while 30.4% retains $e > 0.95$. The near-unity eccentricity tail is therefore measurement-sensitive, but the broader radial-orbit claim is robust.

Table 16. Adaptive Monte Carlo orbit propagation for the Tier A+B+C catalogue. Each star is sampled from the full Gaia DR3 five-parameter astrometric covariance, an independent radial-velocity error, and the Bailer-Jones+2021 photogeometric distance posterior; samples flagged near the threshold are refined to 5,000 realisations and a 100-star convergence sub-sample is re-run at 10,000.

Quantity	Value
N stars (Tier A+B+C)	632
Pass-1 realisations / star	500
Refinement realisations / star (560 stars)	5,000
Convergence sub-sample (100 stars)	10,000
Median per-star MC median e	0.958
Median per-star MC 16th-pct e	0.930
Fraction with MC 16th-pct $e > 0.80$ (%)	99.2
Fraction with MC 16th-pct $e > 0.95$ (%)	30.4
Convergence: max $\lvert\Delta R_{\rm peri,p50}\rvert$ (pc)	8.0
Convergence: max $\lvert\Delta e_{\rm p50}\rvert$	0.0021

Halo-mass sensitivity on the full Tier A+B+C catalogue.—A full-sample halo-mass sensitivity study rescales the Hunter+2024 NFW halo by $\pm15\%$ for all 632 Tier A+B+C stars. Across that range, median eccentricity remains 0.960–0.962, the $e > 0.95$ fraction remains 67.0–67.7%, and median $R_{\rm peri}$ varies only from 145 to 156 pc. The high-eccentricity result is driven primarily by the observed phase-space configuration rather than by the adopted halo mass.

Bar-perturbation sensitivity for a stratified compact-orbit subset.—Because many catalogue stars reach the inner Galaxy, we also integrate a barred Hunter/Sormani variant with pattern-speed sensitivity at $\Omega_p = 33$, 37.5, and $41\,\mathrm{km\,s^{-1}}$ kpc $^{-1}$. Barred integrations leave the qualitative low-$V_{\rm grf}$ radiality intact but change the central reach and bridger counts. The static potential yields zero bridgers, while the barred default yields 10; across the tested pattern speeds the bridger count spans 0–10. This 333-star stratified compact-orbit subsample brackets the commonly adopted Milky Way bar pattern-speed range around 35–$45\,\mathrm{km\,s^{-1}\,kpc^{-1}}$; it is used only as a systematic test of inner-Galaxy geometry, not as a replacement for the static-potential catalogue values.

Radial-velocity uncertainty robustness.—As an additional defensive cut, we required Gaia DR3 radial-velocity uncertainty $\sigma_{RV} < 2\,\mathrm{km\,s^{-1}}$ on top of the Gold filter. This removes the noisier RVS solutions and retains 1,042 of 2,859 stars (36.4%); the comparatively aggressive reduction reflects that DR3 RVS uncertainties are intrinsically larger than $2\,\mathrm{km\,s^{-1}}$ for many giants at the typical sample distance. Within the surviving subset the median eccentricity is essentially unchanged (0.977 vs. 0.975 for the full Gold sample), the median $V_{\rm grf}$ is unchanged ($19.76\,\mathrm{km\,s^{-1}}$ vs. $19.81\,\mathrm{km\,s^{-1}}$), and 100% retain $e > 0.8$. The high-eccentricity claim survives this additional quality cut.

Table 17. Galactic-potential sensitivity. The Hunter+2024 axisymmetric model is the default; the halo NFW component is rescaled by ±15% in mass to bracket plausible literature ranges (e.g. Cautun et al. 2020; Wang et al. 2020). Median values are taken over the Tier A+B+C sample ($N = 632$). The orbital geometry is stable across this halo-mass range.

Halo variant	Median e	$e > 0.95$ (%)	Median $R_{\rm peri}$ (pc)	Median $R_{\rm apo}$ (kpc)
$0.85 \times M_{\rm halo,default}$	0.960	67.7	156	8.08
default ($M_{\rm halo} = 1.27 \times 10^{12}\,M_\odot$)	0.961	67.4	151	8.08
$1.15 \times M_{\rm halo,default}$	0.962	67.0	145	8.08

4.6. *Energy and Angular-Momentum Distribution*

A standard Gaia-era diagnostic is the orbital energy versus z-component of angular momentum (E–L_z); the same diagnostic underlies the substructure surveys of Naidu et al. (2020) and the proto-Galaxy mapping of Rix et al. (2022). In the static Hunter+2024 potential, the Tier A+B+C catalogue occupies low angular momentum and strongly bound inner-Galaxy phase space. This supports the same catalogue-first interpretation used throughout the paper: the sample overlaps familiar Gaia-era accretion and in-situ regions, but the present work does not assign progenitor membership from E–L_z alone. The per-star catalogue reports E, L_z, and Stäckel-fudge actions for follow-up clustering work.

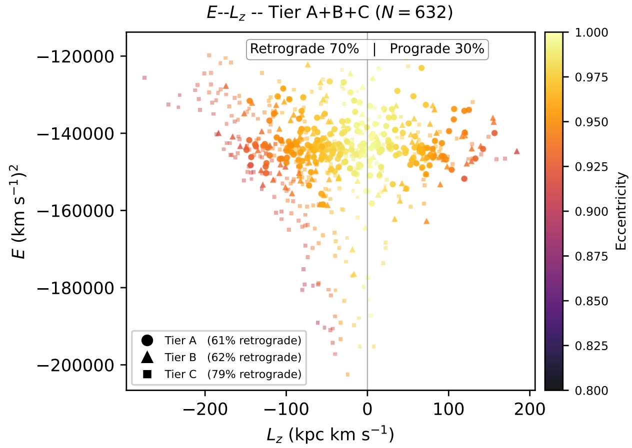

Figure 13. Tier A+B+C slow catalogue ($N = 632$) in E–L_z space, colour-coded by orbital eccentricity in the static Hunter+2024 potential. Marker shape encodes tier (circle/triangle/square = A/B/C); the vertical line marks $L_z = 0$ and the header annotation reports the 70%/30% retrograde/prograde split. The Tier A+B subset is 61–63% retrograde while Tier C is 79% retrograde, the population shift discussed in §5.

4.7. *Extreme Radial-Reach Tail*

The extreme radial-reach tail is now reported in Section 4.4. In the default static Hunter+2024 potential, no Tier A+B+C star satisfies the bridger criterion ($R_{\rm peri} < 2\,\mathrm{kpc}$ and $R_{\rm apo} > 15\,\mathrm{kpc}$). The barred sensitivity model produces 10 bridgers, with a 0–10 range across tested bar pattern speeds. We therefore do not treat the bridger count as a robust population fraction. Likewise, dedicated Monte Carlo propagation of the four Sgr A* candidates yields zero stars with a $> 50\%$ probability of approaching within 10 pc, though three pass the softer 100 pc criterion.

5. INTERPRETATION AND LIMITS

The strongest statement in this paper is empirical. Gaia DR3 contains a tiered catalogue of stars observed at very low Galactic rest-frame speed: 334 stars at Tier A+B confidence and 632 stars at Tier A+B+C confidence. The exact $V_{\rm grf} = 25\,\mathrm{km\,s^{-1}}$ boundary is less secure for individual near-threshold sources, so the catalogue is explicitly probabilistic rather than a single hard-cut list.

A velocity-stratified control comparison across four bands shows that median pericentre grows smoothly with Galactic rest-frame speed in the adopted static potential. This continuous trend shows that the small-pericentre character of the slow-star catalogue is the low-velocity endpoint of a broader kinematic sequence in the

Table 18. Solar-parameter sensitivity of the Tier-A/B/C catalogue counts and median orbital geometry. The four variants span the literature-quoted range of R_0, $z_\odot$, V_c, and the LSR motion (GRAVITY Collaboration 2022; Schönrich 2012; Reid & Brunthaler 2020). Tier counts vary by $\sim$ 50% across the variants, but ensemble orbital geometry is stable to $\sim$ 12% in $R_{\rm peri}$ and $\sim$ 1% in $R_{\rm apo}$ and eccentricity.

Variant	R_0 (kpc)	V_c ($\mathrm{km\,s^{-1}}$)	$N_{\rm A}$	$N_{\rm B}$	$N_{\rm C}$	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)
default	8.178	232.0	215	116	300	150	8.10
GRAVITY+22	8.275	229.0	234	132	355	154	8.14
LSR-6 (S12)	8.178	232.0	249	156	365	141	7.99
RB20	8.178	248.5	176	100	239	163	8.10

Table 19. Sensitivity battery for the Tier A+B+C catalogue ($N = 632$). Median orbital geometry is computed in the Hunter+2024 potential at default solar parameters and varied along three orthogonal axes: NFW halo mass, bar pattern speed, and distance estimator. The sample is stable to $\lesssim$ 15% across each axis; counts and bridger statistics are most sensitive to bar pattern speed.

Axis	Variant	Med. $R_{\rm peri}$ (pc)	Med. $R_{\rm apo}$ (kpc)	Med. e	N bridgers
Halo mass	$0.85 \times M_{\rm halo}$	156	8.08	0.960	—
Halo mass	default ($1.27 \times 10^{12}\,M_\odot$)	151	8.08	0.961	—
Halo mass	$1.15 \times M_{\rm halo}$	145	8.08	0.962	—
Bar pattern speed	$\Omega_p = 33\ \mathrm{km\,s^{-1}\ kpc^{-1}}$	16	8.7	0.996	0
Bar pattern speed	$\Omega_p = 37.5$ (default)	19	9.4	0.996	10
Bar pattern speed	$\Omega_p = 41$	17	9.4	0.996	7
Distance estimator	inverse parallax (no ZP)	64	5.21	0.974	24
Distance estimator	inverse parallax (ZP-corrected)	138	8.04	0.961	5
Distance estimator	Bailer-Jones photogeometric (default)	151	8.08	0.961	0

Note—The bar-pattern-speed test uses a 333-star stratified sub-sample at $\Delta t = 0.1\,\mathrm{Myr}$ trajectory sampling. The distance-estimator test demonstrates that the Lindegren+2021 zero-point and Bailer-Jones photogeometric prior shift orbital geometry consistent with the well-known Lutz-Kelker bias of inverse-parallax distances at low parallax-SNR.

observed Gaia DR3 6D catalogue, not a special physical boundary at exactly $25\,\mathrm{km\,s^{-1}}$.

The dwell-time analysis (Section 4.3) and the matched-control gradient (Section 4.2) together make a single physical statement. A kinematic cut at very low $V_{\rm grf}$ selects stars on inner-reaching, radially biased orbits during a low-speed phase. This is related to the same orbital-turning-point mechanism discussed in the accretion-debris literature, but the present catalogue does not identify the selected stars with GS/E or any other progenitor.

Monte Carlo orbit propagation provides the uncertainty characterisation for the main dynamical claim. At the 16th percentile of each star's posterior, 99.2% of the Tier A+B+C catalogue remains above $e > 0.8$, while 30.4% remains above $e > 0.95$. The strongly radial character of the sample is robust; the exact size of the near-unity eccentricity tail is more sensitive to measurement uncertainties.

The Tier A+B+C catalogue is also dominantly retrograde: 442 of 632 stars (69.9% ± 1.8%) have $L_z < 0$ in the static Hunter+2024 potential. The retrograde fraction climbs across tiers, from 61.2% ± 3.3% in Tier A through 62.5% ± 4.4% in Tier B to 79.2% ± 2.4% in Tier C. Splitting each tier at its own median $V_{\rm grf}$ Monte Carlo posterior width gives 58.9%/63.6% (Tier A narrow/wide), 58.3%/66.7% (Tier B), and 75.8%/82.6% (Tier C); the climb is present in both halves and survives quartile and energy-tertile robustness checks, so it is a property of tier membership rather than an artefact of $V_{\rm grf}$-posterior width. We read the climb as a real population effect: Tier C, the slice nearest the $V_{\rm grf} = 25\,\mathrm{km\,s^{-1}}$ cut, draws more deeply on the accreted-halo phase-space density than Tier A. This dynamical retrograde majority is not reflected in the α-classified subset, where 71% of the 63 stars sit in Splash- or disc-like literature regions. We attribute the mismatch to spectroscopic targeting bias—APOGEE and GALAH are weighted toward brighter, more metal-rich stars—rather than to a re-interpretation of the catalogue's dynamical character. The retrograde majority is consistent with literature accreted-halo populations (Belokurov et al. 2018; Helmi et al. 2018; Naidu et al. 2020), but a population assignment is beyond the scope of this catalogue paper.

As a final present-day compactness check, we searched the slow catalogue for open-cluster- or association-like aggregates in 3D Galactocentric position and velocity (Table 20). No Tier A+B pair lies within 25 pc, and the closest Tier A+B pair is separated by 38.9 pc. The result is unchanged for the broader Tier A+B+C sample and for all 709 point-estimate $V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}$ stars: no pair lies within 25 pc, and no density-based DBSCAN/FoF search (Huchra & Geller 1982; Ester et al. 1996; Castro-Ginard et al. 2020; Hunt & Reffert 2023) with open-cluster-like position–velocity thresholds identifies a three-member compact group. Thus the catalogue is not explained by one or more present-day bound clusters or compact associations. This does not rule out common accretion-related origins or action-space substructure, which need not remain compact in present-day position.

The spectroscopic cross-match sharpens the physical picture while stopping short of an origin claim. The alpha-classified subset contains only 63 Tier A+B+C stars, so population-region fractions are quoted only inside that subset. The sample overlaps phase-space regions often discussed in the Gaia-era accretion literature (Belokurov et al. 2018; Helmi et al. 2018), but this paper is not a membership or merger-decomposition analysis. Because the sample is selected on instantaneous state, not on origin, chemistry, action space, or a fully forward-modelled selection function, we do not attempt progenitor assignment here.

The principal open questions are interpretive rather than catalogue-level. The Gaia DR3 RVS selection function has been applied as a coarse population-scale correction, but its prior-mean fallback affects many slow-star colour–magnitude cells. Spectroscopic coverage is incomplete. Very small inferred pericentres and bridger counts are model-sensitive. The sample is not volume-complete. These limitations set the boundary of the present paper's claims.

6. CONCLUSIONS

We present a tiered Gaia DR3 catalogue of stars with very low Galactic rest-frame speeds. The headline Tier A+B catalogue contains 334 stars with $P(V_{\rm grf} < 25\,\mathrm{km\,s^{-1}}) > 0.84$; the Tier A high-confidence core contains 214 stars with $P > 0.95$, and the broader Tier A+B+C catalogue contains 632 stars with $P > 0.50$.

In the default static Hunter+2024 potential, the Tier A+B+C catalogue maps onto strongly radially biased model orbits with median eccentricity 0.961, median $R_{\rm peri} = 162\,\mathrm{pc}$, and median $R_{\rm apo} = 8.08\,\mathrm{kpc}$. Adaptive Monte Carlo orbit propagation confirms robustness at the $e > 0.8$ level: 99.2% of stars retain $e > 0.8$ at the 16th MC percentile.

Selection-function weighting is reported as context rather than as a precise deprojection. The full Gaia DR3 RVS inverse-selection weighted effective Tier A+B count is 539, while a conservative bracket that gives prior-mean cells unit weight gives 398.

Table 20. Present-day compact-group search for the low-$V_{\rm grf}$ catalogue. Separations are computed in 3D Galactocentric position using the adopted default distances and Solar frame.

Sample	N	Closest pair (pc)	Pairs < 25 pc	Compact groups
Tier A+B	334	38.9	0	0
Tier A+B+C	632	38.9	0	0
Point-estimate $V_{\rm grf} < 25\,{\rm km\,s^{-1}}$	709	32.2	0	0

Note. Compact groups are DBSCAN/FoF detections with at least three members under open-cluster-like position–velocity thresholds (< 25 pc and $< 5\,{\rm km\,s^{-1}}$), plus more permissive checks out to 100 pc and $10\,{\rm km\,s^{-1}}$ for at least five members. No configured search finds a candidate group.

Chemistry is subset-qualified. Only 63 Tier A+B+C stars have both [Fe/H] and alpha-element measurements sufficient for literature-region comparisons, so Splash, Gaia-Sausage/Enceladus, Aurora, and disc-like fractions are quoted only within that alpha-classified subset.

The barred-potential sensitivity test produces 10 bridgers under the default bar pattern speed and 0–10 across the tested 333-star stratified compact-orbit subsample; the static default produces zero. Four Sgr A* candidates were evaluated with dedicated Monte Carlo propagation and interpolated minimum-finding: none have $P(r_{\rm sph,min} < 10\,{\rm pc}) > 0.5$, while three have $P(r_{\rm sph,min} < 100\,{\rm pc}) > 0.5$.

The natural next catalogue step is Gaia DR4: applying the same probabilistic low-$V_{\rm grf}$ selection to the longer-baseline release will test the stability of the present DR3 sample and enlarge the accessible low-speed catalogue.

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Data Availability.—The observational source catalogues used in this work are publicly available from the respective survey archives (Gaia DR3 via the ESA Gaia Archive; APOGEE DR17 via the SDSS DR17 archive; GALAH DR3 via the GALAH Survey website; the Queiroz et al. 2023 StarHorse value-added catalogue via VizieR; and the Zhang et al. 2023 Gaia XP parameter catalogue via the GAVO Heidelberg TAP service as `xpparams.main`). The propagated candidate pool, the Tier A, Tier A+B, Tier A+B+C, and Tier D catalogues, the four matched-control band catalogues, the per-star Monte Carlo results, and the orbit-derived columns are staged for deposition in a persistent repository before journal submission, with the repository DOI cited in the submitted manuscript. The staged release includes the FITS catalogues, control catalogues, validation products, Gaia Archive selection query, pipeline scripts, potential configuration files, and environment metadata needed to reproduce the numerical products from the released intermediate tables.

Software Availability.—The review-stage analysis pipeline, including sample construction, control-band matching, orbit integration, Monte Carlo propagation, and the figure- and table-generation scripts, is available in a version-controlled GitHub repository at https://github.com/wodvik/gaia-slow-vgrf-catalogue. The repository also contains the review-stage catalogue products, column documentation, environment file, configuration file, Gaia Archive selection query, and the current manuscript draft. After final review, the frozen software and data-release versions will be archived in Zenodo and the resulting persistent DOI will be cited in the submitted manuscript.

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